

Does Idiomatic Phrasing Degrade LLM Task Performance?

A Controlled Study Across Instruction-Tuned Decoder Models

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Submitted as final project report for the NLP course, Reichman University, 2026

1 Introduction

Idiomatic expressions — those whose meaning cannot be derived compositionally from their individual words, e.g., “*spill the beans*” or “*par for the course*” — are routinely treated in NLP as a harder case for language models than direct, literal phrasing. The premise is intuitive: idioms are culturally specific and non-compositional, so a model must have learned an expression rather than derived it, and models do degrade on expressions they have not seen [1]. It is, however, rarely tested *in isolation*: any two rewordings of a sentence already differ lexically and syntactically, so an observed accuracy drop after “idiomatizing” it could simply be an artifact of rewording in general.

Central research question: does phrasing a task input idiomatically cause a measurable degradation in a language model’s task accuracy, *over and above* the degradation that any comparable paraphrase (a non-idiomatic rewording) already causes?

To answer this cleanly, we rewrite every original example x into two controlled variants that share the same rewriting budget but differ in exactly one property: $x_{\text{paraphrase}}$, reworded **without** idioms, the *control* condition isolating the effect of rewording alone; and $x_{\text{idiomatic}}$, reworded **with** one or more natural idiomatic expressions and the same meaning. The label y is given to the rewriter and re-checked afterwards, failures being dropped rather than relabelled (§2.2), and each variant is evaluated under an identical prompt template, so only the input’s surface form changes between runs. The question matters practically wherever informal text reaches a classifier rather than being authored for one — review and social-media sentiment, support-ticket triage, content moderation — because there figurative phrasing is a property of the incoming data rather than a choice the system can make; SST-2’s own snippets are already partly figurative before any rewriting (§2.3). If idiomaticity has an isolable negative effect, that bears on prompt engineering, augmentation strategy, and robustness evaluation; if apparent idiom failures are merely generic paraphrase sensitivity, effort spent specifically on idiom handling is misplaced relative to general robustness-to-rewording work.

Contributions. We contribute (i) the paraphrase-controlled design described above, which isolates the idiom-specific effect from the generic cost of rewording — instantiating for a linguistic property the control-condition logic concurrent work establishes in general [2], and finding a residual that survives its baseline on inference, unlike that demographic case; (ii) a *sole-cause* diagnostic that, restricted to items a model already answers correctly, identifies which single rewrite is responsible when exactly one of the two flips the answer; (iii) dual significance testing, pairing a continuity-corrected McNemar test with a non-parametric bootstrap under different assumptions, guarding against the chi-square approximation on small discordant-pair counts; (iv) breadth — 14 instruction-tuned decoder LLMs from 0.5B to 7B across 10 families, on three datasets spanning sentiment, knowledge QA and inference, for 42 runs with a pooled cross-model test per dataset; and (v) a reproducible four-stage pipeline with LLM-judge validation of every augmented example and per-artifact provenance (§2).

1.1 Related Work

Idiom processing in NLP. Prior work has mostly focused on *detecting* or *translating* idioms rather than measuring their effect on downstream accuracy. Tayyar Madabushi et al. [1] probe pretrained models on the literal-versus-idiomatic distinction, finding they do reasonably well given one or a few labelled examples per expression but degrade substantially zero-shot on unseen expressions, and Chakrabarty et al. [3] introduce FLUTE, a 9,000-instance figurative-language benchmark spanning idioms, similes, metaphors, and sarcasm. The LLM era has largely inherited this framing: Phelps et al. [4] find prompted LLMs competitive but still short

of fine-tuned task-specific models on three idiomaticity datasets even at GPT-4 scale, and Mi et al. [5] show with a controlled contrastive dataset that LLMs frequently fail to resolve idiomaticity when doing so depends on context, arguing that headline benchmark success may rest on reasoning shortcuts. All of this measures idiom *comprehension* — establishing it as a genuine weak point — rather than what idiomatic phrasing costs on a task that is not itself about idioms.

Figurative language inside a task. Closer to our question is work that scores figurative material within a downstream task. Stowe et al. [6] pair idiomatic and metaphoric sentences with literal counterparts and find that MNLI-finetuned RoBERTa reliably recognizes figurative–literal entailment yet fails on matched non-entailing pairs. For metaphor, Sanchez-Bayona and Agerri [7] conclude from NLI and QA experiments that LLM performance is shaped more by lexical overlap and sentence length than by metaphorical content — an independent finding consonant with ours. Our design departs from these in introducing the figurative property by a controlled rewrite of items whose original form is also scored, rather than annotating it where it already occurs, which is what makes a matched control arm possible.

Why a paraphrase control is necessary. Models are sensitive to input edits that leave the correct answer intact: Ribeiro et al. [8] flip predictions by *rewording* inputs under semantics-preserving rules (SEARs), and for LLMs the effect is larger still — Cao et al. [9] report a 45.5-point gap between the best and worst semantically equivalent phrasings of the same input. A bare original-versus-idiomatic contrast is therefore uninterpretable on its own, because it measures idiomaticity confounded with generic rewording sensitivity.

Controlled evaluation design. Both our rewrites are invariance tests in the sense of CheckList [10] — label-preserving perturbations under which the prediction ought not to change. They are deliberately the mirror image of contrast sets [11], which perturb minimally in order to *change* the gold label. What we add is subtraction: reporting the gap between a treatment and a matched control condition instead of the treatment alone, the move that control tasks make for probing classifiers [12]. Control conditions have appeared in figurative evaluation before — MUNCH includes deliberately inapt paraphrases to detect lexical-similarity shortcutting [13] — but as a distractor in the answer space, not as a matched arm whose accuracy is subtracted. Concurrent work argues the general case directly: Yang et al. [2] observe that any counterfactual edit bundles the factor of interest with incidental surface variation, and report that on MedQA a 14.9% flip rate under a demographic edit is statistically indistinguishable from the 14.1% induced by paraphrasing alone. We take that logic as settled and instantiate it for a linguistic intervention; unlike the demographic case, a residual effect does survive here on inference. Studies that inject a property without such a baseline, such as the dialect rewrites of GLUE by Ziems et al. [14], accordingly bound its contribution from above rather than estimating it. To our knowledge idiom-inserted rewrites of general-purpose NLU benchmarks have not previously been scored against a matched, semantics-preserving paraphrase arm over the same items, which is what licenses attributing the residual to idiomaticity rather than to rewording.

2 Methodology

2.1 Pipeline overview

The project is implemented as four independently re-runnable stages (Figure 1) that communicate only through versioned files on disk, plus YAML configs that set each stage’s run parameters; dataset-specific loading and answer-parsing logic lives in per-dataset modules. Any stage can therefore be re-executed alone — evaluating a new model does not require repeating cleaning or augmentation — and every artifact records the config hash and model/prompt version that produced it.

Stage 1: Clean. Each raw dataset is loaded from Hugging Face and reduced to one clean CSV of rows suitable for rewriting: rows outside a configured length band are dropped (very short inputs leave no room for an idiom; very long ones are expensive to augment and dilute the measured effect), text is normalized (Unicode NFC, whitespace collapsing), exact duplicates are removed, and every retained row gets a stable identifier that aligns all its downstream variants. The label y and dataset-specific metadata (e.g., MMLU’s answer choices) carry through unchanged.

2.2 Augmentation into paraphrase and idiomatic variants

Stage 2: Augment. Each cleaned row is rewritten by an external LLM (Google Gemini) into two variants using two distinct, fixed prompt templates. One Gemini sub-model is pinned per dataset for cost reasons (`gemini-3.6-flash` for SST-2 and MMLU, `gemini-3.5-flash-lite` for MNLI), shared by both arms, so the paraphrase control stays internally valid within a dataset; only magnitudes *across* datasets are not cleanly

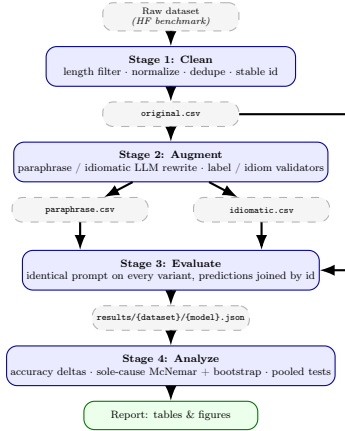


Figure 1: The four-stage pipeline (the stage orchestrators carry no dataset- or model-specific branching; each dataset adds its own loader and evaluator module). The same pipeline instantiates once per dataset/model combination. Raw data is cleaned (Stage 1), rewritten into validated paraphrase and idiomatic variants (Stage 2), evaluated per model across every variant with an identical prompt (Stage 3), and aggregated into the accuracy-delta and sole-cause significance analysis (Stage 4). `original.csv` feeds both Stage 2 (as the rewriting source) and Stage 3 (as the control variant) directly.

Table 1: One aligned *sole-cause* triple per dataset, verbatim from `datasets_out/`: the original and paraphrase were answered correctly and the idiomatic variant was not, so the idiom alone broke the answer. Only one field per dataset is rewritten — SST-2’s sentence, MMLU’s question stem, MNLI’s premise — everything else is byte-identical across rows. **Bold** marks the aligned span each variant puts in the same role. The MNLI idiom pairs a negation with *sell short*; the SST-2 idiom reads as positive inside a negative-sentiment snippet.

MNLI — hypothesis: “ <i>No one would give them credit like that.</i> ”; gold label <i>contradiction</i> . Idiomatic-only-wrong for 12 of 14 models.	
original	Even the Scots are likely to give them more credit than that.
paraphrase	Even the residents of Scotland will probably accord them a higher level of recognition than that amount.
idiomatic	Even the Scots are unlikely to sell them short like that.
SST-2 — gold label <i>negative</i> . Idiomatic-only-wrong for 10 of 14 models.	
original	in which the rest of the cast was outshined by ll cool j.
paraphrase	in which the rest of the cast was outperformed by ll cool j.
idiomatic	in which LL Cool J stole the show from the rest of the cast.
MMLU — gold answer “ <i>The nation invests in research and development of new technology.</i> ”; the four choices are untouched. Idiomatic-only-wrong for 8 of 14 models.	
original	Which of the following scenarios would increase a nation’s production possibility frontier (PPF)?
paraphrase	Which of the following situations would expand a country’s production possibility frontier (PPF)?
idiomatic	Which of the following scenarios would push out a nation’s production possibility frontier (PPF)?

comparable. Templates are selected per dataset so wording suits the input type (MNLI, whose rewrite target is a premise judged against a fixed hypothesis, uses its own NLI-specific pair):

the *paraphrase prompt* asks for a rewording in different words that explicitly avoids idiomatic expressions, and the *idiomatic prompt* asks for one that naturally incorporates one or more idioms or figures of speech while preserving meaning, register, and length. Both are given the task label y in the prompt itself, so a rewrite cannot silently invert the correct answer, e.g., flipping a positive sentiment snippet into a negative one. An external, deliberately stronger LLM is used rather than one of the models under evaluation, since the augments’s own idiom/paraphrase judgment must be more reliable than the systems being tested.

Every augmented row then passes a validator chain: **label preservation**, an LLM-judge check that the rewrite has not altered the correct label; and **idiom presence/absence**, a per-variant LLM-judge check requiring a genuine idiom in the idiomatic variant and none in the paraphrase. Rows that fail validation after a bounded number of retries are dropped from that variant (recorded in a sidecar manifest for transparency), which is why the paraphrase/idiomatic CSVs are typically somewhat smaller than the original CSV. An on-disk response cache keyed by (prompt hash, augments model, input id) makes reruns idempotent and avoids re-incurring API cost. Table 1 shows one produced triple per dataset.

2.3 Datasets

Three datasets are evaluated, each scored with the accuracy metric used in its original publication. Stage 2 rewrites one field per dataset — SST-2’s sentence, MMLU’s question stem, MNLI’s premise — and everything else, including MMLU’s answer choices and MNLI’s hypothesis, stays byte-identical across the three variants.

SST-2 [15] is binary sentiment classification over short movie-review snippets; **MMLU** [16] is multiple-choice knowledge questions spanning 57 subjects; and **MNLI** [17] is three-way inference (*entailment* / *neutral* / *contradiction*) over premise-hypothesis pairs from ten written and spoken genres.

The evaluated split differs by dataset according to what each release makes available with labels: MMLU uses its **validation** split and MNLI its two validation splits (below), while SST-2, whose GLUE test labels are withheld, draws from **train** and **validation** combined, capped at 2,000 rows.

The three cover qualitatively different task types under one protocol: short, informal, already-somewhat-figurative text; longer, formal, fact-oriented multiple-choice questions; and sentence-pair inference. MNLI is the most direct test of the three, because in sentiment or knowledge QA an idiom perturbs the surface of an input whose answer is otherwise independent of it, whereas an entailment judgement turns on a literal reading of the premise itself. Its three-way label space also sets chance at 33% rather than 50%, which depresses the per-row rate of sole-cause events relative to a binary task.

MNLI-specific protocol. MNLI pairs roughly three hypotheses with every premise; emitting all of them would repeat the same premise and violate the independence assumption behind the paired test, so the loader keeps only the first hypothesis per premise. Rows come from the two validation splits combined (matched, 9,815 pairs; mismatched, 9,832), the test labels being withheld. The mismatched half contributes the five genres absent from training; each row’s genre is recorded, but the analysis does not split on it. After first-hypothesis selection and cleaning, 5,127 premises survive to form the MNLI original file, of which 4,709 (91.8%) additionally clear validation in *both* rewritten variants and so form the aligned triples the analysis runs on.

2.4 Evaluated models

Rather than a single-model case study, we evaluate a broad panel of 14 open, instruction-tuned decoder LLMs spanning 0.5B to 7B parameters across 10 distinct model families and pretraining lineages: the Qwen2.5-Instruct family (0.5B / 1.5B / 7B), the Falcon3-Instruct family (3B / 7B), the OLMo-2-Instruct family (1B / 7B), Gemma-2-2B-it, Mistral-7B-Instruct-v0.3, Phi-4-mini-instruct (3.8B), Granite-3.1-2B-Instruct, H2O-Danube3-4B-Chat, Yi-1.5-6B-Chat, and SmolLM2-1.7B-Instruct. Tables and prose below drop the **-Instruct/-it/-Chat** suffix. All models are evaluated zero-shot with an identical dataset-specific prompt template applied to all three variants of a row, so only the input text differs between runs; a device-agnostic runner selects CUDA, Apple-Silicon MPS, or CPU and adjusts precision accordingly.

Stage 3: Evaluate. For each (model, dataset) pair, the model is run on all three variant CSVs with the identical prompt, and predictions are joined by row id, yielding one matched (original, paraphrase, idiomatic) triple of predictions per surviving row. Per-variant accuracy is computed directly from these joined predictions, scoring as incorrect any generation the dataset’s parser cannot map to a label — a choice §3 re-tests on the subset where all three variants parsed.

2.5 Statistical framework

Stage 4: Analyze. Two families of statistic are computed. The first is per-variant accuracy and the two deltas between variants, reported throughout in percentage points (pp):

$$\Delta_{\text{par}} = \text{acc}_{\text{par}} - \text{acc}_{\text{orig}}, \quad \Delta_{\text{idi}} = \text{acc}_{\text{idi}} - \text{acc}_{\text{par}}. \quad (1)$$

Δ_{par} measures the cost of rewording and Δ_{idi} the additional cost of idiomaticity on top of it; the research question of §1 is precisely whether $\Delta_{\text{idi}} < 0$. Each is accompanied by a paired-bootstrap confidence interval and a McNemar exact p-value over the corresponding discordant pairs.

The second, and our primary diagnostic, is a *sole-cause* test, which attributes degradation to a single rewrite type rather than to ordinary noise. It restricts attention to rows where all three variants were produced and the model answered the **original** correctly — so any subsequent error is attributable to the rewrite, not to the item being hard for that model — and then counts $b = \textit{idiomatic-only-wrong}$ (the idiomatic rewrite alone broke an otherwise-correct answer) and $c = \textit{paraphrase-only-wrong}$ (the paraphrase rewrite alone broke it). Rows where both rewrites agree, breaking it or not, carry no information about which is worse and are excluded, exactly as in any paired test. The gap between b and c is then assessed two ways:

Table 2: Per-dataset summary over aligned triples. Δ = mean pp across models; *sig.* = models with significant negative/positive Δ_{idi} ($p < 0.05$, McNemar exact). *Parse-clean* repeats both after dropping triples with any parse failure. Pooled sole-cause sums b (idiomatic-only-wrong) and c (paraphrase-only-wrong) across models, tested by continuity-corrected McNemar and a 10,000-resample bootstrap.

Dataset	Models	n	All triples			Parse-clean		Pooled sole-cause				
			Δ_{par}	Δ_{idi}	sig.	Δ_{idi}	sig.	b / c	χ^2	p_{χ^2}	p_{boot}	
SST-2	14	1,548	-1.28	+1.35	0 / 5	+0.76	0 / 3	533 / 723	28.4	<0.0001	<0.0001	
MMLU	14	1,085	+0.13	-0.79	1 / 0	-0.77	1 / 0	474 / 378	10.6	0.0011	<0.0001	
MNLI	14	4,709	-0.68	-1.28	10 / 0	-1.26	10 / 0	2,804 / 2,246	61.4	<0.0001	<0.0001	

Table 3: Per-model Δ_{idi} (pp), ordered by parameter count, two panels. **Bold:** Δ_{idi} itself significant ($p < 0.05$, McNemar exact). $\dagger$: the separate sole-cause b vs. c gap is significant on *both* McNemar and bootstrap (either direction) — independent of bold, so a cell may carry either, both, or neither. Negative values support the hypothesis; SST-2 is almost entirely positive.

Model	SST-2	MMLU	MNLI	Model	SST-2	MMLU	MNLI
Qwen2.5-0.5B	+0.45	-1.75	-0.40	Phi-4-mini-3.8B	-0.19	-1.84	-1.51 $\dagger$
OLMo-2-1B	+1.29	-0.28	-1.70 $\dagger$	H2O-Danube3-4B	+1.74	-2.49	+0.04
Qwen2.5-1.5B	+1.16	-2.03 $\dagger$	-1.59 $\dagger$	Yi-1.5-6B	+0.97	-0.18	-1.89 $\dagger$
SmolLM2-1.7B	+2.20	-0.74	+0.30	Qwen2.5-7B	+3.04 $\dagger$	+0.09	-1.17 $\dagger$
Gemma-2-2B	+1.36 $\dagger$	-1.38	-1.51 $\dagger$	Falcon3-7B	-0.26	+0.65	-1.23 $\dagger$
Granite-3.1-2B	+1.49 $\dagger$	+2.12	-0.89	Mistral-7B-v0.3	+4.13 $\dagger$	-1.20	-2.08
Falcon3-3B	+0.39	-0.37	-2.25 $\dagger$	OLMo-2-7B	+1.16	-1.66 $\dagger$	-2.04 $\dagger$

McNemar’s test [18], the standard test for paired nominal before/after comparisons, with the continuity correction of Edwards [19]:

$$\chi^2 = \frac{(|b - c| - 1)^2}{b + c}, \quad \text{df} = 1, \quad (2)$$

testing the null hypothesis that b and c are drawn from the same underlying 50/50 rate (i.e., among discordant rows, it is a coin flip which rewrite broke it).

Non-parametric bootstrap check. Because that chi-square approximation can be unreliable when $b + c$ is small, every result is cross-checked with a paired bootstrap [20]: the aligned rows are resampled with replacement 10,000 times, $b - c$ is recomputed on each resample, and a 95% percentile interval plus a two-sided empirical p-value are read off the result. A result counts as robustly significant only when both checks agree, as they do in all 42 per-model runs.

The same sole-cause statistics are additionally pooled across all evaluated models for a given dataset (summing b and c , and resampling from the concatenated pooled rows for the bootstrap), with the explicit caveat that this pooling is not a rigorous mixed-effects test, since different models are not strictly exchangeable independent draws, and is reported as indicative rather than definitive.

3 Experimental results

The sweep covers 42 (model, dataset) runs — all 14 models on each of the three datasets — 126 variant-level inference passes, about 33 hours of accelerator time. Every number below is computed over *aligned triples*: rows where the original and both validated rewrites exist and parsed, leaving 1,548 / 1,085 / 4,709 rows for SST-2 / MMLU / MNLI (Stage 3 also scores the full original files — 2,000 / 1,107 / 5,127 rows — which have no counterpart to pair against).

One manipulation, three directions (Table 2). The same idiomatize-a-matched-paraphrase intervention produces a significant *penalty* on MNLI (10/14 models, pooled sole-cause $p < 10^{-4}$), a significant *benefit* on SST-2 (5/14 positive, none negative, pooled sole-cause significant in the opposite direction), and a marginal penalty on MMLU that survives only in aggregate (1/14 individually, $p = 0.0011$ pooled). Idiomaticity is therefore not a general accuracy tax; its sign and size track how directly the rewritten span carries the decision.

MNLI: the configuration the design predicts should show the effect. Ten of 14 models show a significant penalty, from Falcon3-3B at -2.25 pp to Qwen2.5-7B at -1.17 pp (Table 3), and the sole-cause test agrees — $b > c$ for 12 of 14, both tests concurring at $p < 0.05$ for 9 of the same 10 significant models. Only Mistral-7B-v0.3 splits: a significant accuracy delta with a non-significant sole-cause gap ($b = 225$, $c = 191$). Nine models agreeing on two tests with different evidence bases is the strongest single result in this study, and it lands exactly where §2

Table 4: Worked disagreement cases: three rows where a model with a *significant* Δ_{idi} predicted differently under the paraphrase and the idiomatic rewrite (✓ correct, × wrong). **Bold** marks the aligned span each variant puts in the same role. Every variant is scored under the same fixed system prompt (one word/letter answer per dataset) and only the rewritten field changes within a block. The MNLI and MMLU blocks show the majority direction, where the idiom alone breaks a correct answer; the SST-2 block shows that dataset’s reversal, where the paraphrase breaks the answer and the idiom restores it.

Variant	User prompt	Prediction
MNLI — Falcon3-3B-Instruct ($\Delta_{\text{idi}} = -2.25$ pp). Every prompt ends Hypothesis: <i>She now knows what was going on in her mind.</i> / Answer: . Gold: <i>entailment</i> .		
original	Premise: It took me really a long time to sort what was going on in my mind out , she said.	entailment ✓
paraphrase	Premise: She stated that it required a significant amount of time for her to organize her thoughts .	entailment ✓
idiomatic	Premise: It took me quite a while to wrap my head around what was going on in my mind, she said.	contradiction ×
SST-2 — Mistral-7B-Instruct-v0.3 ($\Delta_{\text{idi}} = +4.13$ pp). Gold: <i>positive</i> .		
original	Sentence: is n’t embarrassed to make you reach for the tissues	positive ✓
paraphrase	Sentence: is not ashamed to make you cry	negative ×
idiomatic	Sentence: isn’t shy about pulling at your heartstrings and making you reach for the tissues	positive ✓
MMLU — H2O-Danube3-4B-Chat ($\Delta_{\text{idi}} = -2.49$ pp). The four choices are untouched: A. a logical mistake. B. an ethical mistake. C. a factual mistake. D. not a mistake. Gold: <i>D</i> .		
original	Question: According to Nagel , the view that moral luck is paradoxical is:	D ✓
paraphrase	Question: Based on Nagel’s analysis , the perspective that moral luck is a paradox is:	D ✓
idiomatic	Question: In Nagel’s eyes , the view that moral luck is paradoxical is:	A ×

predicted it should: entailment turns on a literal reading of the premise, so a figurative surface adds a resolution step that can fail. Table 4 works one such row through in full for a significant model on each dataset.

SST-2: the idiom recovers what the paraphrase cost. Original accuracy is high and tight (83.8–91.7%); plain paraphrasing already costs 12 of 14 models, significantly for 4, and idiomatizing that same rewording recovers most of the loss — positive for 12 of 14, significant for 5, with Mistral-7B-v0.3 swinging widest (−2.91 pp then +4.13 pp). This is partly genuine: SST-2’s idioms tend to carry sentiment rather than obscure it (Table 1). It is partly artifact, addressed below.

MMLU: a small effect capped by what gets rewritten. MMLU spreads models much more widely (29.7–68.7%, against a 25% floor) while both deltas stay small (+0.13/−0.79 pp); only H2O-Danube3-4B reaches individual significance, but the sole-cause direction is consistent enough ($b > c$ for 9 of 14) that pooling turns it significant. This is likely structural: Stage 2 rewrites only the question stem, leaving the answer choices — where an MMLU item’s actual knowledge requirement sits — untouched, so an idiom can obscure what is asked without touching what the model must know. Models near the chance floor also generate too few sole-cause events to have power regardless (OLMo-2-1B: $b = 8$, $c = 11$).

The SST-2 reversal is partly a parsing artifact. Unparseable generations score as incorrect, so a variant provoking more format violations is penalised for reasons unrelated to idiom comprehension. Failure rates are low on average (1.0%/0.2%/0.3% for SST-2/MMLU/MNLI) but uneven — up to 6.7% for Mistral-7B-v0.3 on SST-2 — and there the *paraphrase* arm fails more often than the idiomatic arm for 11 of 14 models, inflating Δ_{idi} . Dropping every triple with any parse failure leaves MMLU and MNLI untouched (all 10 significant MNLI models survive) but cuts SST-2’s mean from +1.35 to +0.76 pp and its significant count from 5 to 3; Qwen2.5-7B is the clearest case, its apparent +3.04 pp advantage becoming −0.21 pp once its 6.4%-vs-2.1% parse-failure gap is removed. The direction survives regardless — even filtered, no SST-2 model shows a significant idiom *penalty*, only a smaller advantage.

No capability or scale trend. Δ_{idi} correlates with neither original accuracy (−0.13/+0.28/−0.29) nor parameter count (+0.10/+0.24/−0.30) at any significance ($p > 0.3$ throughout). The Qwen2.5 ladder makes the point concretely: MNLI’s penalty is non-monotone across its three sizes (−0.40, −1.59, −1.17 pp at 0.5B/1.5B/7B), and all four 7B models in the panel sit inside the degraded group rather than above it. Rewriting is not purely destructive, either — pooled across models, 419 SST-2, 235 MMLU and 1,614 MNLI rows were wrong on the original but right on both rewrites.

4 Discussion

The control arm is what makes the data legible. Score idiomatic directly against original — the design most idiom studies use — and SST-2 looks like a null (+0.07 pp, 1/14 significant) while MNLI’s cost is overstated by roughly 53% (the naive -1.96 pp bundles in -0.68 pp of plain rewording). The matched arm turns that false null into two real, cancelling effects on SST-2 and corrects MNLI to its true size; neither correction is visible without it.

Where the effect lives matters more than whether it exists. Idiomaticity costs accuracy exactly when the figurative surface sits on the span the answer is computed from: MNLI rewrites the premise a literal reading depends on (-1.28 pp); MMLU rewrites a stem while the knowledge sits in untouched choices (-0.79 pp); SST-2’s idioms carry the very property being classified, so they help rather than hurt ($+1.35$ pp). This is a ranking, not a magnitude comparison — MNLI’s augments and rewrite depth differ from the other two — but the ordering matches exactly what a load-bearing-span account predicts, and it is a narrower, more falsifiable claim than “idioms degrade LLMs.” It tempers without contradicting the comprehension weaknesses Tayyar Madabushi et al. [1] and Chakrabarty et al. [3] report directly: what we measure is the residual cost after a model applies whatever partial idiom understanding it has, on a task it was never asked to reason about idioms for.

The magnitudes are real but modest, and scale is not a fix. A 1–2 pp MNLI drop is statistically robust but the same order as plain rewording’s own cost, not an order above it — idiomaticity is one distinguishable contributor to paraphrase sensitivity, not a separate failure mode. No correlation with scale exceeds $|\rho| = 0.30$ across a 14-fold parameter range, but with only 14 models $|\rho| \gtrsim 0.53$ is the significance floor: absence of evidence, not evidence of absence. The practical upshot is a triage rule, not a scaling law — invest in idiom robustness where figurative language can land on the decision-bearing span, not where a bigger checkpoint is assumed to absorb it.

4.1 Threats to validity

The statistical treatment has three known weaknesses. The parse-failure filter is blunt — it drops 2.2% of SST-2 rows on average but 12.4% for Mistral-7B-v0.3 there, and 8.2% for OLMo-2-1B on MNLI, and dropping rows non-randomly can itself bias a comparison — so filtered and unfiltered readings bracket the truth. The pooled tests also treat models’ items as exchangeable when the same row recurs once per model, so a row that is idiom-hard for many models is resampled as many independent observations — clustering that narrows pooled intervals and makes pooled p-values optimistic. No mixed-effects model was run, so pooled figures — including MMLU’s, which rests entirely on pooling — are indicative, not definitive. No multiple-comparisons correction is applied either: 14 tests at $\alpha = 0.05$ expect 0.7 hits under a true null, so MMLU’s single one carries little weight ($P(X \geq 1) = 0.51$) while MNLI’s 10 and SST-2’s five positives are clear of chance (8×10^{-11} , 4×10^{-4}).

Scope limits. Every model was evaluated zero-shot, greedily, under a single prompt template per dataset, so few-shot prompting, sampled decoding and prompt-format variation are untested and could interact with idiomaticity; the panel is also decoder-only.

4.2 Future work

The most valuable next step would sharpen the MNLI finding rather than broaden it: a direct check that rewrites preserve meaning, ruling out semantic drift as an alternative explanation. A mixed-effects re-analysis would then put the pooled claims on a proper footing, and extending to few-shot prompting would establish whether in-context examples recover the loss, as labelled examples do for the fine-tuned encoders of Tayyar Madabushi et al. [1].

Three further extensions would test *why* the effect concentrates where it does rather than only *that* it does.

Idiom opacity is currently binary. Scoring each inserted idiom’s compositional transparency — via the augments itself, or existing psycholinguistic norms — and correlating it against Δ_{idi} would test whether cost scales with non-compositionality rather than mere idiom presence. This reuses data already collected, with no new model runs required.

The same logic applies to frequency: idiom rarity, via corpus statistics or target-model perplexity, would test the familiarity account of Tayyar Madabushi et al. [1] directly rather than by analogy. Like the opacity check, this is a post-hoc correlation against the existing 42 runs. A dose-response relationship here would strengthen the mechanistic story behind the observed penalty.

A third extension would turn the load-bearing explanation above from a cross-dataset correlation into a manipu-

lated variable, by rewriting MNLI’s hypothesis (or MMLU’s answer choices) in addition to the premise/stem so idiom position varies within one dataset rather than across three. Unlike the first two, this requires new augmented data and fresh evaluation runs.

5 Summary

We measured the accuracy cost of idiomatic phrasing for 14 instruction-tuned decoder LLMs from 0.5B to 7B parameters on SST-2, MMLU and MNLI, pairing every idiomatic rewrite with a matched non-idiomatic paraphrase of the same row so that the cost of rewording is subtracted rather than absorbed into the effect. That control is what makes the result readable. Without it, SST-2 looks like a null and MNLI’s idiom-specific penalty is overstated by roughly 53%. With it, the penalty is confined to the configuration where the rewritten span is the one the label is computed from: -1.28 pp on MNLI, significant for 10 of the 14 models, against -0.79 pp on MMLU with a single significant model and no penalty at all on binary sentiment. Idiomaticity is therefore a distinguishable contributor to paraphrase sensitivity rather than a separate failure mode, and no scale trend is detectable across a 14-fold parameter range, though 14 models could not detect a moderate one.

6 Code

Code repository: https://github.com/LihuZur/Idiom_degradation

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